

Project Report: Diabetes Hospital Readmission Analysis

Data Analysis Midterm Project | Malak Usama Mohamed

1. Dataset Overview

This project uses the *Diabetes 130-US Hospitals for Years 1999–2008* dataset, sourced from the UCI Machine Learning Repository and originally published by Strack et al. (2014) using the Health Facts clinical database (Cerner Corporation). The dataset contains 101,766 real, de-identified inpatient encounters from 130 U.S. hospitals, covering patients with a diabetes-related diagnosis. For this project, the original 50 attributes were narrowed to 22 core variables relevant to the business objective — demographics, admission/discharge details, clinical utilization metrics, diabetes-specific test results and medication indicators, and the readmission outcome — with three additional descriptive label columns engineered from coded ID fields.

Business domain: Healthcare — hospital operations and clinical quality analytics.

Objective: identify which patient and encounter-level factors are most strongly associated with 30-day hospital readmission in diabetic patients, and translate those patterns into a KPI-driven dashboard that hospital administrators could use to target high-risk patients and monitor performance against a readmission benchmark.

Why this dataset: it reflects genuine hospital operations data rather than a synthetically generated dataset, which gives variable relationships real analytical credibility. The healthcare domain also aligns with my professional background working with CMS Medicare claims and insurance data, which supported interpretation of the variables and framing of the business questions.

2. Data Cleaning

Data quality issues identified:

- 2,273 rows (2.2%) in **race** used a “?” placeholder instead of a proper missing-value marker.
- 3 rows in **gender** were coded “Unknown/Invalid.”
- **admission_type_id**, **discharge_disposition_id**, and **admission_source_id** were stored only as numeric codes, with no human-readable label — a data type/usability issue for reporting.
- **max_glu_serum** and **A1Cresult** were roughly 95% and 83% “None,” though this reflects a genuine clinical category (test not ordered) rather than a data error.
- 30,248 encounters belonged to patients with more than one visit in the dataset — a real repeat-patient pattern, not a duplicate-record error (no fully duplicate rows or duplicate encounter IDs were found).
- No negative values or impossible entries were found in any numeric field.

Cleaning techniques performed:

- Recoded the 2,273 “?” values in **race** to “Prefer Not to Say.”
- Retained the 3 “Unknown/Invalid” gender rows (recoded to “Prefer Not to Say”) rather than removing them, to avoid discarding real data for an assignment that does not require perfect category purity.
- Built lookup/mapping tables in Power Query to translate **admission_type_id**, **discharge_disposition_id**, and **admission_source_id** into readable labels, merged in alongside the original IDs so both are available for traceability.

- Deliberately kept **max_glu_serum** and **A1Cresult** in their original categories rather than dropping them, since “not tested” is a meaningful clinical signal relevant to the analysis.
- Confirmed and documented, rather than removed, expired/deceased discharge dispositions; these are excluded specifically from readmission-rate calculations at the analysis stage rather than from the raw dataset.
- Verified correct data types for all numeric and categorical fields during Power Query load.

Remaining limitations:

Readmission counts above roughly 7 prior inpatient visits are based on very small sample sizes and produce unstable percentages (e.g., swinging to 0% or 100%); these are treated as unreliable and excluded from the headline dashboard visual rather than removed from the dataset. Additionally, patient-level repeat encounters were kept in the dataset as independent rows, which means observations are not fully statistically independent — acceptable for descriptive dashboard analysis, but a limitation if inferential statistics were to be applied.

3. Exploratory Data Analysis

- **Prior inpatient visits are the strongest predictor of readmission.** The 30-day readmission rate rises from 8.29% for patients with zero prior inpatient visits to 34.95% at six prior visits — roughly a four-fold increase. Beyond ~7 prior visits, sample sizes become too small to trust (percentages swing erratically), so this pattern is reported only up to the reliable range.
- **Elevated glucose readings show a clear readmission signal.** Patients with a glucose serum test result above 300 show a 16.55% 30-day readmission rate, compared to 10.9–12.7% for all other test categories.
- **Insulin dosage changes are associated with higher readmission.** Patients whose insulin dose was increased (12.62%) or decreased (13.72%) show meaningfully higher 30-day readmission than those with no change (9.90%) or steady dosing (10.63%), suggesting dose instability may signal harder-to-manage cases.
- **Age shows a steady, intuitive trend.** Readmission climbs from 1.58% in the 0–10 age group to 11.86% in the 80–90 group, before a slight dip in the 90–100 group.
- **Race and admission type show modest, less decisive differences.** Caucasian patients show the highest 30-day rate (11.23%) and Asian patients the lowest (9.42%); Emergency admissions (11.36%) outpace Elective (10.10%), while the Trauma Center category’s 0% rate reflects an extremely small sample rather than a real effect.
- **A1C testing shows a weak, inconsistent relationship** with readmission compared to the glucose test — notably, patients with no A1C test recorded show a slightly higher 30-day rate (11.25%) than those tested and found normal (9.48%), a pattern opposite to what was seen for glucose. This is reported as a genuine null/weak finding rather than omitted.
- **Number of medications and length of stay are moderately correlated** ($r = 0.47$), indicating longer stays are associated with more complex medication regimens, though the relationship is far from deterministic.

4. Dashboard Summary

The Power BI dashboard is organized into three pages, each answering a distinct layer of the business question.

Page 1 — Executive Overview: a KPI gauge for 30-day readmission rate against a target benchmark, summary cards for total encounters and average length of stay, and bar charts breaking down readmission rate by age group and admission type, with slicers for age, race, admission type, and gender.

Page 2 — Clinical Risk Factors: the dashboard's core analytical story — a line chart of readmission rate by number of prior inpatient visits, column charts for glucose test result and insulin dosage change, and a bubble scatter chart showing the relationship between number of medications and length of stay, sized by encounter count.

Page 3 — Demographics & Testing Patterns: readmission rate by race and by A1C test result, alongside a pivot table cross-tabulating age group against readmission status, with slicers for race and A1C result.

Together, the three pages move from a single top-line performance number, into the clinical drivers behind it, and finally into demographic and testing context — directly answering which patients are highest risk and why.

5. KPI Explanation

KPI: 30-Day Readmission Rate — the percentage of encounters resulting in hospital readmission within 30 days of discharge.

Target: 10%, based on the general 30-day all-cause readmission benchmark commonly referenced under CMS's Hospital Readmissions Reduction Program (HRRP), which is widely used across the industry as a reasonable performance line for hospital quality.

Why this target is appropriate: it reflects a real, externally grounded quality standard rather than an arbitrary number, and sits close enough to this dataset's overall rate to create a meaningful, discussable comparison rather than a target that is trivially met or hopelessly out of reach.

Result: the overall 30-day readmission rate in this dataset is 10.99%, essentially at the target line. However, this aggregate figure masks substantial variation: patients with several prior inpatient admissions, or with glucose readings above 300, readmit at rates two to three times higher than the target, while lower-risk segments (younger patients, no prior admissions, elective admissions) sit comfortably below it. This indicates that overall performance is borderline, but risk is highly concentrated in identifiable subgroups — a more actionable insight than the aggregate number alone.

6. Business Insights and Recommendations

- **Target patients with 3+ prior inpatient visits for proactive discharge planning.** This group already shows readmission rates roughly double the overall average, and utilization history is available at admission — making it a practical, immediately actionable risk flag.
- **Prioritize glucose monitoring protocols for patients with serum glucose above 300.** This subgroup shows the sharpest single-category jump in readmission risk in the dataset, suggesting tighter inpatient glycemic management could have an outsized impact on reducing readmissions.
- **Review insulin dose-adjustment practices.** Since both increases and decreases in insulin dosage correlate with higher readmission, this may reflect harder-to-stabilize patients rather than the dosage change itself causing harm — worth a clinical follow-up study rather than an immediate policy change, since this analysis cannot establish causation.
- **Reconsider A1C testing's role in inpatient protocol given its weak signal here.** While glucose testing shows a strong relationship with readmission, A1C testing does not show a consistent one in this data, suggesting hospital resources may be better spent on other risk markers identified above.
- **Use the KPI dashboard to monitor subgroup performance, not just the aggregate rate.** Since the overall 10.99% rate sits right at the proposed 10% target, tracking only the aggregate could mask high-risk segments that are performing far worse; segment-level tracking (by prior admissions and glucose result) is recommended going forward.

Note: this analysis is observational and descriptive. Relationships identified (e.g., insulin dose changes, A1C testing) indicate association, not causation, and should inform further clinical investigation rather than be treated as proven cause-and-effect drivers.